A Detail Injection-Based Fusion Framework for Hyperspectral, Multispectral, and Panchromatic Remote Sensing Images

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***Abstract*—The primary goal of fusing hyperspectral image (HSI), multispectral image (MSI), and panchromatic (PAN) images is to transfer spatial details from MSI and PAN into HSI to generate high-resolution HSIs (HR-HSIs). Nevertheless, most existing methods rely on black-box deep learning (DL) architectures and rarely leverage this fundamental principle, resulting in limited physical interpretability. To address this challenge, we introduce a novel detail injection-based model for hyperspectral, multispectral, and panchromatic imagery (DIM-HMPF). The framework is engineered to enrich the spatial detail of HSI through fusion with MSI and PAN modalities, all while maintaining high-fidelity spectral information. Therefore, we utilize the physical model of detail injection for unfolding, using DL to learn this process. We further employ an optimization problem-solving method based on the proximal gradient descent (PGD) algorithm, achieved through a series of iterative steps. By unfolding these iterative steps into several network modules, DIM-HMPF is ultimately obtained. On the representative remote sensing hyperspectral datasets of Chikusei, Houston, and Xiongan, both quantitative evaluation metrics and visual comparison results consistently demonstrate that DIM-HMPF significantly outperforms existing mainstream fusion algorithms in terms of reconstruction accuracy, especially as evidenced by its superior performance in structural preservation and spectral fidelity. These results comprehensively validate the distinct superiority, robustness, and effectiveness of DIM-HMPF within the HMPF framework, as well as its strong generalization capability and practical potential in complex scenarios. Our code is publicly available at https://github.com/Freelancefangjian/DIM-HMPF**

***Index Terms*—Hyperspectral image (HSI), multispectral image (MSI), panchromatic image.**

I. Introduction

HYPERSPECTRAL imaging [1], [2] is a rapidly advancing technology that captures hyperspectral images (HSIs), consisting of hundreds of contiguous spectral bands. It has a wide range of applications in computer vision and remote sensing [3], including change detection [4], [5], [6], object identification [7], object segmentation [8], environmental monitoring [9], and agricultural analysis [10]. However, due to physical limitations in sensor design, there is an inherent tradeoff between spatial and spectral resolution. While HSIs offer rich spectral detail, their spatial resolution tends to be low because each band receives limited energy. In contrast, multispectral image (MSI) [11] or panchromatic (PAN) [12] images feature fewer bands but significantly higher spatial resolution. To address this disparity, HSI fusion techniques combine the spectral richness of HSIs with the fine spatial detail of MSI or PAN images acquired over the same area, effectively producing high-resolution HSIs (HR-HSIs).

Leveraging the strengths of deep convolutional neural networks (CNNs) [13], [14], [15], [16], [17], researchers have increasingly focused on deep learning (DL)-driven HSI fusion methods [18], [19], [20], [21]. Contemporary mainstream CNN architectures generally adopt a dual-input paradigm of HSI–MSI fusion (HMF) being a typical example. For instance, Xie et al. [18] introduced MHF-Net, which incorporates the image acquisition model of low-resolution inputs and exploits the low-rank nature of HR-HSIs in the spectral domain, realized through an iterative optimization algorithm based on the proximal gradient method. More recently, Dong et al. [21] developed an iterative HSI super-resolution (SR) approach built upon a deep HSI denoiser. Although HMF methods generally preserve high spectral fidelity, their capacity to enhance spatial resolution is inherently constrained by the spatial limits of MSI. This limitation has prompted a shift in recent efforts toward PAN-HSI fusion, leveraging PAN, which offers significantly higher spatial resolution than MSI. However, the stark spectral and spatial disparities between HSI and PAN data often result in severe spectral distortion, rendering the fusion task highly ill-posed.

Fusing HSI–MSI–PAN data [22], [23], [24], [25], [26] provides an effective way to generate imagery with both high spatial resolution and rich spectral information. By combining the complementary characteristics of PAN (providing fine spatial details), MSI (bridging spectral gaps), and HSI (ensuring spectral fidelity), the limitations of single-modality fusion can be effectively overcome. However, most existing HSI–MSI–PAN fusion methods do not explicitly follow the core physical principle of this task, namely, injecting spatial details extracted from MSI and PAN images into

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low-resolution HSI to reconstruct HR-HSIs. The prevailing DL-based approaches [22], [25] typically rely on black-box end-to-end mappings, in which the physical roles of different modalities during the fusion process are implicitly learned through network parameters rather than explicitly modeled. Although some studies [24] attempt to characterize the imaging process using degradation models within a deep unfolding framework, their focus remains on degradation-reconstruction consistency, and the intrinsic connection to the detail injection mechanism has not been sufficiently explored. More importantly, traditional detail injection models [27] have achieved long-standing and mature success in pan-sharpening and HSI–MSI fusion tasks, demonstrating strong physical interpretability and robustness. Nevertheless, in the more complex HSI–MSI–PAN fusion scenario, there is still a lack of DL-based methods that can systematically incorporate and effectively exploit this physically meaningful detail injection mechanism.

Based on the above analysis, this article revisits traditional detail injection models and reformulates them within a unified optimization framework, incorporating algorithm unfolding techniques to propose a heuristic detail injection model for HSI–MSI–PAN fusion, as illustrated in Fig. 1. This model explicitly embeds the physical process of spatial detail injection into a learnable network architecture, thereby enhancing physical interpretability while enabling effective integration of model-based priors and data-driven learning. Specifically, a method is first developed to extract spatial details from MSI and PAN images, based on which an objective optimization model is constructed with three detail fidelity terms to characterize the injection of multisource details into HSI. The optimization problem is then solved via iterative gradient descent, and the iterative process is unfolded into an end-to-end deep network, allowing the key parameters to be adaptively learned from data rather than relying on fixed values as in traditional algorithms. Furthermore, to fully exploit the complex nonlinear complementary features among HSI, MSI, and PAN, CNNs are employed to replace the function transformation modules in the optimization model. Unlike conventional black-box deep networks, the proposed network is explicitly guided by the optimization model, endowing each network variable with a clear physical meaning and thereby significantly improving interpretability.

The following summarizes the main contributions of this article.

1) This article proposes a physically interpretable deep fusion framework that incorporates traditional detail injection models into a unified optimization formulation. By integrating algorithm unfolding techniques, the spatial detail injection mechanism is explicitly embedded into a learnable network, enabling effective synergy between model-based priors and data-driven learning.
2) Within this fusion framework, a multisource detail constraint and end-to-end learning approach is designed. An optimization model with three detail fidelity terms is constructed to characterize the injection of spatial details from MSI and PAN into HSI, and the iterative solution is unfolded into an end-to-end deep network, allowing the model parameters to be adaptively learned from data.
3) The nonlinear complementary features among HSI, MSI, and PAN are explored by replacing the function transformation modules in the optimization model with CNNs. Guided by the optimization framework, the network architecture fully captures these complex nonlinear complementary features, while ensuring that the network variables retain explicit physical meaning, thereby enhancing both interpretability and fusion performance.

The structure of the remaining sections of this article is as follows. Section II introduces the background relevant to our study, while Section III offers a comprehensive description of our proposed detail injection-based model for hyperspectral, multispectral, and panchromatic imagery (DIM-HMPF). This includes details on image registration methods, the rationale behind the model, network implementation, and proximity operations. Section IV presents the experimental results along with a detailed discussion, while Section V concludes the study and provides insights into future directions.

II. Related Works

The multisource image fusion techniques designed to enhance LR-HSI can be broadly classified into three main categories: HSI–PAN fusion, HSI–MSI fusion, and HSI–MSI–PAN fusion.

A. HSI–Pan Fusion

Panchromatic–hyperspectral fusion (often referred to as hyperspectral pansharpening) enhances the spatial resolution of LR-HSIs by injecting the fine spatial detail from an HR-PAN image—adapting methodologies originally established in multispectral fusion. These methods fall into four principal categories: component substitution (CS) [28], [29], [30], which substitutes a low-frequency component of the HSI (via transformations like IHS or PCA) with spatial content from the PAN image; multiresolution analysis (MRA) [31], which utilizes multiscale tools such as wavelet or Laplacian pyramid transforms to extract spatial features from the PAN image and inject them into the hyperspectral data; variational optimization (VO) [32] (or more broadly, model-based optimization), which frames pansharpening as an ill-posed inverse problem and resolves it via an energy minimization framework; and DL approaches, which leverage neural network architectures—such as HyperPNN [33], which employs spectrally predictive convolutions to better preserve spectral fidelity, MFT-GAN [34], a multiscale feature-guided transformer network optimized for unsupervised hyperspectral pansharpening, and HyperGAN, a GAN-based model incorporating attention mechanisms for richer spectral–spatial fusion. Crucially, the pronounced difference in resolution between single-band PAN imagery and multiband hyperspectral imagery presents a persistent challenge in maintaining spectral fidelity during the fusion process.

B. HSI–MSI Fusion

In recent years, HSI and MSI image fusion has become a crucial technique for generating HR-HSIs. Existing research

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on HSI–MSI fusion can be broadly divided into two categories: traditional model-based approaches and DL-based data-driven approaches. Traditional model-based methods typically rely on the physical observation mechanism of hyperspectral imaging. By incorporating priors such as sparse representation [35], [36], [37], [38] and low-rank constraints [39], [40], [41], [42], these methods formulate an optimization problem and employ iterative solvers to recover high-quality spectral and spatial details. Owing to their clear physical interpretability and solid theoretical foundation, these methods have long been dominant. However, they are often sensitive to noise, degradation model mismatch, and the complexity of real-world data, which may significantly degrade their performance in complex scenarios. With the rapid development of DL, data-driven fusion methods have gradually become the mainstream research direction. Many studies utilize CNNs [43], [44], [45], [46] to learn end-to-end mappings from MSI and low-resolution HSI to HR-HSI. At the same time, generative adversarial networks (GANs) [47], [48] have been introduced to enhance spatial textures and structural information through adversarial learning. More recently, Transformer architectures [20], [49], [50] have attracted attention for their strong capability in modeling long-range dependencies and cross-modal relationships, enabling more effective capture of complex spectral–spatial correlations and improving cross-modal complementarity. Furthermore, some studies attempt to combine the advantages of both model-driven and data-driven approaches. A key direction in this regard is algorithm unrolling, which unfolds traditional iterative optimization steps into trainable deep networks while maintaining strong physical interpretability. Representative examples include MHF-net [18], MoG-DCN [21], and EDIP-Net [51], all of which incorporate imaging degradation models, physical priors, or deep image priors into network design to achieve improved fusion performance. To further reduce reliance on large-scale paired training datasets—which are often costly and difficult to obtain—researchers have proposed unsupervised methods [44], [52], [53], zero-shot learning approaches [54], and deep image prior-based techniques [51]. These strategies aim to enhance the generalization capability, reconstruction accuracy, and computational efficiency of fusion networks, enabling them to better adapt to diverse real-world remote sensing scenarios.

C. HSI–MSI–PAN Fusion

In the study of enhancing the spatial resolution of HSIs, traditional methods typically rely on fusing HSIs with high-resolution MSIs (HR-MSIs). However, the effectiveness of such approaches is inherently limited by the spatial resolution of the MSIs themselves. To overcome this limitation, Li et al. [22] proposed a DL-based fusion network, HyperNet, which simultaneously integrates HSI, MSI, and panchromatic (PAN) images. By injecting spatial details from MSI and PAN while preserving the spectral integrity of the HSI, HyperNet effectively improves the spatial quality of the fused image. To address differences in swath widths among data with varying resolutions, the work in [23] designed a multibranch fusion architecture based on an encoder–decoder framework, incorporating an enhanced hyperspectral encoder and a multilevel spatial–spectral aggregation module, enabling efficient integration of multimodal information within a unified framework. Tian et al. [24] proposed HMPNet, which models the complex relationship between HR-HSI and PAN using a deep prior, achieving interpretable and high-performance fusion without relying on handcrafted priors. To further handle scale inconsistencies, Meng et al. [25] introduced MGFEI-Net, employing an attention-driven hybrid-scale integration module to capture spatial details across multiple resolutions; similarly, Meng et al. [26] developed a spatial–spectral feature transfer and joint reconstruction framework, comprehensively considering scale differences among HSI, MSI, and PAN, and achieving high-fidelity reconstruction in both spatial and spectral domains. Nevertheless, existing methods largely overlook the fundamental principle of HSI–MSI–PAN fusion—namely, injecting spatial details from MSI and PAN into HSI to generate HR-HSIs. Moreover, DL approaches that explicitly leverage this physical mechanism have not been systematically explored, limiting both interpretability and potential performance.

III. Methodology

A. Detail Injection Model

The traditional CS method, as a widely used PAN sharpening algorithm, can be generalized in the following form:

$$\mathbf{X}_n = \mathbf{H}_n + g_n(\mathbf{P} - \mathbf{P}_L) \tag{1}$$

where the subscript $n = 1, 2, \ldots, N$ represents the nth spectral band, $\mathbf{X}_n \in \mathbb{R}^{H\times W}$ and $\mathbf{H}_n \in \mathbb{R}^{H\times W}$, respectively, represent the HR-HSI and the upsampled LR-HSI. g_n represents the detail injection gain factor for each band. $\mathbf{P} \in \mathbb{R}^{H\times W}$ represents the PAN, and the matrix $\mathbf{P}_L \in \mathbb{R}^{H\times W}$ represents the average brightness image calculated from the individual spectral band images of the HSI

$$\mathbf{P}_L = \sum_{n=1}^{N} \omega_n \mathbf{H}_n \tag{2}$$

where the weight vector ω_n is used to measure the spectral overlap between the LR-HSI and PAN channels. The core difference among different CS methods lies in how the weight vector and detail injection gain factors are selected. For instance, a joint optimization estimation method for spectral response weight parameters and detail injection gains has been proposed in the literature, known as the band-dependent spectral detail (BDSD) injection model. The fundamental idea of this method is to compute different band-correlated average brightness images for each spectral band; thus, the format of its detail injection is

$$\mathbf{X}_n = \mathbf{H}_n + g_n\left(\mathbf{P} - \sum_{i=1}^{N} \omega_{n,i}\mathbf{H}_i\right) \tag{3}$$

where $\omega_{n,i}$ consists of N^2 unknown parameters, which can be regarded as an extension of the weight vector in (2).

We notice that the HSI has N bands, while the MSI has B bands; and a band in the MSI may cover a specific spectral

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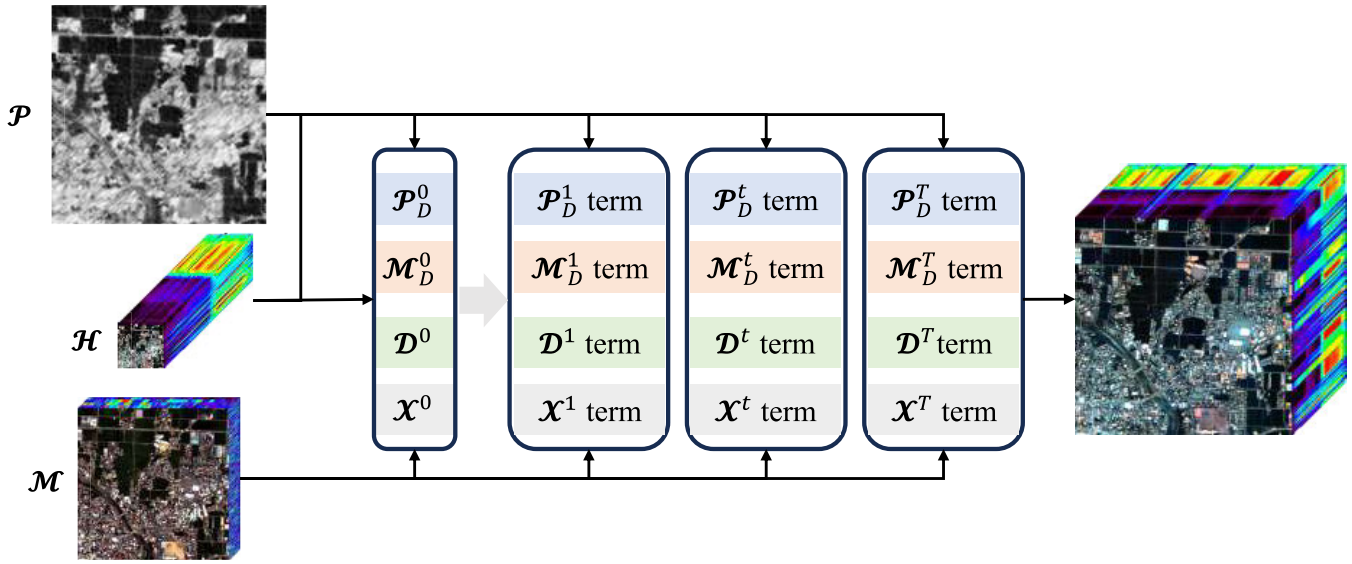


Fig. 1. Illustrates the overall flowchart of the proposed method.

range in the HSI, so it is necessary to specify the MS band image that needs to be fused with the HS bands. Assuming HR-HSI $\mathbf{X}$ can model the sum of the upsampled LR-HSI $\mathbf{H}$ and the detail part of the MSI. Furthermore, assuming that the bth MSI band is assigned to fuse with any n bands within the spectral coverage range. Based on the above analysis, we can assume

$$\mathbf{M}_L \approx \sum_{l \in \mathcal{B}(k')} \omega'_{k,l} \mathbf{H}_l, \quad k' \in \{1, 2, \ldots, N\} \tag{4}$$

where $\omega'_{k,l}$ represents the weights calculated based on spectral responses. Therefore, we have

$$\mathbf{X}_k = \mathbf{H}_k + g_k \left(\mathbf{M}_L - \sum_{l \in \mathcal{B}(k)} \omega'_{l,k} \mathbf{H}_l \right). \tag{5}$$

B. Modeling Motivation

The model proposed in this chapter is derived from the traditional detail injection framework. For ease of model definition, in the following text, both PAN and MSIs are represented in tensor form instead of matrix form. Therefore, the two different forms of detail injection in (1) and (5) can be combined into the following unified form:

$$\mathbf{X} = \mathbf{H} + \Upsilon \left(\mathbf{P} - \mathbf{P}_L, \mathbf{M} - \mathbf{M}_L \right) \tag{6}$$

where $\mathbf{X}$ and $\mathbf{H}$ represent the HR-HSI and the LR-HSI, respectively. $\mathbf{P}$ represents the tensor obtained by replicating the PAN image along the spectral dimension. $\mathbf{P}_L$ represents the average brightness component in the CS method, and here we adopt a more general form of the BDSD method; thus, (2) can be rewritten as

$$\mathbf{P}_L = \overline{\mathbf{H}} \times_3 \mathbf{W} \tag{7}$$

where $\mathbf{P}_L$ represents the brightness component of N bands, $\overline{\mathbf{H}}$ denotes the LR-HSI after bicubic interpolation, and $\mathbf{W}$ is the weight matrix corresponding to unknown parameters in the BDSD model. Operation $\times_3$ indicates a mode-3 tensor multiplication, meaning that the brightness components of different bands are obtained linearly from the upsampled MSI. As a result, the residual image between the PAN and this average brightness image is also band-dependent, which can be expressed as

$$\mathbf{P}_D = \mathbf{P} - \mathbf{P}_L \tag{8}$$

where $\mathbf{P}_D$ represents potential band-related spatial details.

Similar to the spatial detail extraction of the PAN, we can rewrite (4) as

$$\mathbf{M}_L = \mathbf{H} \times_3 \mathbf{W}' \tag{9}$$

where $\mathbf{M}_L$ represents the brightness component of N bands. As a result, the residual image between the MSI and this average brightness image is also band-dependent

$$\mathbf{M}_D = \mathbf{M} \times_3 \mathbf{R}^T - \mathbf{M}_L \tag{10}$$

where $\mathbf{R}^T$ represents the transpose of the spectral response function.

In the traditional BDSD model, the extracted band-related spatial details are multiplied by gain coefficients band by band to obtain the injected details. We generalize this into a more reasonable form by introducing a nonlinear transformation function $\Upsilon(\cdot)$ to characterize the relationship between the injected details and the extracted details. The format of detail injection is thus,

$$\mathbf{X} = \mathbf{H} + \Upsilon \left(\mathbf{P}_D, \mathbf{M}_D \right). \tag{11}$$

Therefore, combining the above equations, we can construct the fidelity term in the following form:

$$\begin{aligned} \mathcal{L} \left(\mathbf{P}_D, \mathbf{M}_D, \mathbf{X} \right) = & \frac{1}{2} \left\| \mathbf{P}_D - (\mathbf{P} - \mathbf{P}_L) \right\|_F^2 \\ & + \frac{\alpha}{2} \left\| \mathbf{M}_D - (\mathbf{M} - \mathbf{M}_L) \right\|_F^2 \\ & + \frac{\beta}{2} \left\| \Upsilon \left(\mathbf{P}_D, \mathbf{M}_D \right) + \mathbf{H} - \mathbf{X} \right\|_F^2 \end{aligned} \tag{12}$$

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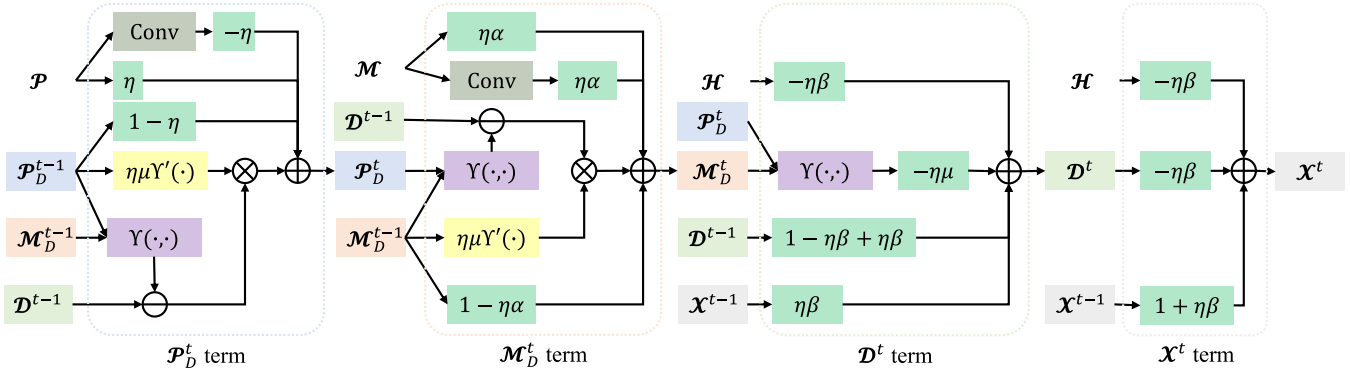


Fig. 2. Illustration of tth iteration phase in DIM-HMPF.

where α and β are the balancing weights of the fidelity term for the MSI and the overall fidelity term, respectively.

Optimize (12)

$$\begin{aligned}\min_{\mathbf{P}_D,\mathbf{M}_D,\mathbf{X}} \quad & \mathcal{L}\left(\mathbf{P}_D,\mathbf{M}_D,\mathbf{X}\right)=\frac{1}{2}\|\mathbf{P}_D-(\mathbf{P}-\mathbf{P}_L)\|_F^2\\ &+\frac{\alpha}{2}\|\mathbf{M}_D-(\mathbf{M}-\mathbf{M}_L)\|_F^2\\ &+\frac{\beta}{2}\|\Upsilon\left(\mathbf{P}_D,\mathbf{M}_D\right)+\mathbf{H}-\mathbf{X}\|_F^2\\ \text{s.t.}\quad & \mathbf{D}=\Upsilon\left(\mathbf{P}_D,\mathbf{M}_D\right).\end{aligned} \tag{13}$$

The half-quadratic splitting (HQS) function of (13) can be expressed as

$$\begin{aligned}\min_{\mathbf{P}_D,\mathbf{M}_D,\mathbf{X},\mathbf{D}} \mathcal{L}\left(\mathbf{P}_D,\mathbf{M}_D,\mathbf{X}\right)=&\frac{1}{2}\|\mathbf{P}_D-(\mathbf{P}-\mathbf{P}_L)\|_F^2\\ &+\frac{\alpha}{2}\|\mathbf{M}_D-(\mathbf{M}-\mathbf{M}_L)\|_F^2\\ &+\frac{\beta}{2}\|\mathbf{D}+\mathbf{H}-\mathbf{X}\|_F^2\\ &+\frac{\mu}{2}\|\mathbf{D}-\Upsilon\left(\mathbf{P}_D,\mathbf{M}_D\right)\|_F^2\end{aligned} \tag{14}$$

where μ is the balancing weights.

C. Model Optimization

Gradient descent can be used in this article to sequentially update the four target variables, and the gradient calculation for each variable is as follows:

$$\begin{cases}\dfrac{\partial\mathcal{L}}{\partial\mathbf{P}_D}=\mathbf{P}_D-(\mathbf{P}-\mathbf{P}_L)\\ \qquad -\mu\Upsilon'\left(\mathbf{P}_D\right)\left(\Upsilon\left(\mathbf{P}_D,\mathbf{M}_D\right)-D\right)\\ \dfrac{\partial\mathcal{L}}{\partial\mathbf{M}_D}=\alpha\left(\mathbf{M}_D-(\mathbf{M}-\mathbf{M}_L)\right)\\ \qquad -\mu\Upsilon'\left(\mathbf{M}_D\right)\left(\Upsilon\left(\mathbf{P}_D,\mathbf{M}_D\right)-D\right)\\ \dfrac{\partial\mathcal{L}}{\partial\mathbf{D}}=\beta\left(\mathbf{D}+\mathbf{H}-\mathbf{X}\right)+\mu\left(\Upsilon\left(\mathbf{P}_D,\mathbf{M}_D\right)-D\right)\\ \dfrac{\partial\mathcal{L}}{\partial\mathbf{X}}=\beta\left(\mathbf{D}+\mathbf{H}-\mathbf{X}\right)\end{cases} \tag{15}$$

where the tth iteration of model optimization can be represented as

$$\begin{cases}\mathbf{P}_D^t=\mathbf{P}_D^{t-1}-\eta\left(\mathbf{P}_D^{t-1}-(\mathbf{P}-\mathbf{P}_L)\right.\\ \quad\left.-\mu\Upsilon'\left(\mathbf{P}_D^t\right)\left(\Upsilon\left(\mathbf{P}_D^{t-1},\mathbf{M}_D^{t-1}\right)-\mathbf{D}^{t-1}\right)\right)\\ \mathbf{M}_D^t=\mathbf{M}_D^{t-1}X-\eta\left(\alpha\left(\mathbf{M}_D^{t-1}-(\mathbf{M}-\mathbf{M}_L)\right)\right.\\ \quad\left.-\mu\Upsilon'\left(\mathbf{M}_D^{t-1}\right)\left(\Upsilon\left(\mathbf{P}_D^{t-1},\mathbf{M}_D^{t-1}\right)-\mathbf{D}^{t-1}\right)\right)\\ \mathbf{D}^t=\mathbf{D}^{t-1}-\eta\left(\beta\left(\mathbf{D}^{t-1}+\mathbf{H}-\mathbf{X}^{t-1}\right)\right.\\ \quad\left.+\mu\left(\Upsilon\left(\mathbf{P}_D^t,\mathbf{M}_D^t\right)-\mathbf{D}^{t-1}\right)\right)\\ \mathbf{X}^t=\mathbf{X}^{t-1}-\eta\beta\left(\mathbf{D}^{t-1}+\mathbf{H}-\mathbf{X}^{t-1}\right)\end{cases} \tag{16}$$

where parameter η is the step size for each iteration, and the superscript indicates the stage of the iteration.

D. Network Design

This article delves into the iterative process for solving the HR-HSI reconstruction problem, cycling repeatedly until the subproblems involving variables $\mathbf{P}_D$, $\mathbf{M}_D$, $\mathbf{D}$, and $\mathbf{X}$ all converge to local minima. Although this iterative approach is theoretically viable, its practical implementation requires significant time and computational resources. To address this challenge, we propose the DIM-HMPF method: by unfolding the iterative steps and employing DL networks to execute each iteration stage, it substantially enhances computational efficiency, as illustrated in Fig. 2. The core of DIM-HMPF lies in its forward computation mechanism, which facilitates HR-HSI inference while implicitly encapsulating the degradation model and data priors. Through this computation, DIM-HMPF progressively reconstructs HR-HSI, thereby improving inference accuracy and robustness. Moreover, DIM-HMPF adopts an end-to-end methodology to optimize the entire reconstruction workflow within a unified framework. Its key strength resides in the autonomous learning of deep priors, degradation model parameters, and hyperparameters—eliminating the need for manual parameter tuning or predefined specifications. This capability effectively enhances the model's generalization performance and adaptability across diverse scenarios.

1) Iterations Unfolding: To streamline network design, this section describes HSI, MSI, and PAN uniformly in tensor format rather than matrix format. This representation

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Algorithm 1 Proposed DIM-HMPF

1: **Input:** HSI $\mathcal{H}$, MSI $\mathcal{M}$, PAN $\mathcal{P}$;
2: **Initialize:** $\mathcal{P}_D^0$, $\mathcal{M}_D^0$, $\mathcal{D}^0$, and $\mathcal{X}^0$ via Eq. (18);
3: **while** $t \leq T$ **do**
4: Update $\mathcal{P}_D^t$ via Eq. (17);
5: Update $\mathcal{M}_D^t$ via Eq. (17);
6: Update $\mathcal{D}^t$ via Eq. (17);
7: Update $\mathcal{X}^t$ via Eq. (17);
8: **end while**
9: **Output:** $\mathcal{X}^T$

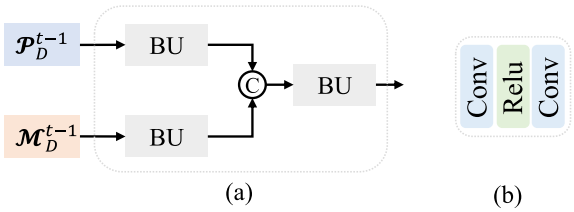


Fig. 3. Architecture of (a) $\Upsilon(\cdot,\cdot)$ and (b) BU.

enables models to process high-dimensional data structures more efficiently, thereby enhancing network flexibility and performance. Based on this representation, consider a HSI $\mathcal{H} \in \mathbb{R}^{H\times W\times N}$, a MSI $\mathcal{M} \in \mathbb{R}^{H\times W\times N}$, and a PAN $\mathcal{P} \in \mathbb{R}^{H\times W\times 1}$. The $\mathbf{P}_D$, $\mathbf{M}_D$, $\mathbf{D}$, and $\mathbf{X}$-subproblems in (16) can be reformulated as tensor expressions as follows:

$$\begin{cases} \mathcal{P}_D^t = \mathcal{P}_D^{t-1} - \eta\left(\mathcal{P}_D^{t-1} - (\mathcal{P} - \mathcal{P}_L)\right. \\ \quad \left. - \mu\Upsilon'\left(\mathcal{P}_D^{t-1}\right)\left(\Upsilon\left(\mathcal{P}_D^{t-1}, \mathcal{M}_D^{t-1}\right) - \mathcal{D}^{t-1}\right)\right) \\ \mathcal{M}_D^t = \mathcal{M}_D^{t-1} - \eta\left(\alpha\left(\mathcal{M}_D^{t-1} - (\mathcal{M} - \mathcal{M}_L)\right)\right. \\ \quad \left. - \mu\Upsilon'\left(\mathcal{M}_D^{t-1}\right)\left(\Upsilon\left(\mathcal{P}_D^{t}, \mathcal{M}_D^{t-1}\right) - \mathcal{D}^{t-1}\right)\right) \\ \mathcal{D}^t = \mathcal{D}^{t-1} - \eta\left(\beta\left(\mathcal{D}^{t-1} + \mathcal{H} \quad - \mathcal{X}^{t-1}\right)\right. \\ \quad \left. + \mu\left(\Upsilon\left(\mathcal{P}_D^{t}, \mathcal{M}_D^{t}\right) - \mathcal{D}^{t-1}\right)\right) \\ \mathcal{X}^t = \mathcal{X}^{t-1} - \eta\beta\left(\mathcal{D}^t + \mathcal{H} - \mathcal{X}^{t-1}\right). \end{cases} \tag{17}$$

In this article, the algorithm unfolding technique is employed to transform each step of Algorithm 1 into corresponding network layers, thereby constructing an interpretable deep network model. Specifically, the tth stage of the proposed DIM-HMPF model corresponds to the tth iteration of Algorithm 1.

The fusion process is illustrated in Fig. 3. It first extracts detailed information using a convolution, followed by a ReLU activation function and another convolution. These extracted details are then integrated through concatenation. Next, the integrated features are further refined using another sequence of convolution, ReLU activation, and convolution, and finally, the fusion result is produced.

Before starting the iterative learning process, the four positional variables are initialized

$$\begin{cases} \mathcal{P}_D^0 = \mathcal{P} - \overline{\mathcal{H}} \\ \mathcal{M}_D^0 = \mathcal{M} \times_3 \mathbf{R}^T - \overline{\mathcal{H}} \\ \mathcal{D}^0 = \left(\mathcal{P}_D^0 + \mathcal{M}_D^0\right)/2 \\ \mathcal{X}^0 = \mathcal{M} \end{cases} \tag{18}$$

where $\mathbf{R}^T$ represents the transpose of the spectral response function, which is replaced by a convolutional layer with kernel size and N feature maps.

E. Loss Function

All parameters in DIM-HMPF are trained end-to-end. The loss function is constructed to include the L1 norm of the reconstruction error between the potential HR-HSI $\hat{\mathbf{X}}$ and the estimated HR-HSI $\mathbf{X}$

$$\text{loss} = \frac{1}{N}\sum_n \left\|\hat{\mathcal{X}}_n - \mathcal{X}_n\right\|_1 \tag{19}$$

where $\hat{\mathcal{X}}_n$ and $\mathcal{X}_n$, respectively, refer to the nth patch of the reference HR-HSI and the estimated HR-HSI.

IV. Experiments

In this section, we systematically describe the experimental setup and performance evaluation of the detail injection-based HMPF network to verify its effectiveness in addressing the HMPF problem. First, the datasets and evaluation metrics adopted in the experiments are introduced. Subsequently, the proposed DIM-HMPF is compared with existing advanced methods. It is worth noting that research on HMPF is still relatively limited, and most existing methods do not release their codes. Therefore, we select a series of representative hyperspectral–multispectral and hyperspectral–panchromatic image fusion methods for comparison. Specifically, the comparison methods include three typical traditional approaches: the polynomial kernel method with 23 coefficients (EXP) [55], the subspace regularization method (HySure) [56], and the FUSE method [57]; methods for hyperspectral and MSI fusion, including the dual U-Net (D-UNet) [58] and the pyramid shuffle-and-reshuffle transformer (PSRT) [59]; as well as methods for hyperspectral and panchromatic image fusion, including the spectrally predictive CNN-based hyperspectral pansharpening method (HyperPNN) [33], the deep residual learning network for boosting hyperspectral pansharpening accuracy (DRCNN) [60], and an interpretable model-driven deep network (HMPNet) [24]. All experiments are conducted on the PyTorch platform using a computer equipped with an NVIDIA RTX 3060 graphics card.

A. Experimental Dataset

In this study, three publicly available HSI datasets were used to evaluate the effectiveness of the proposed method, namely the Chikusei, Houston, and Xiongan datasets. Partial images from the test sets of these three datasets are depicted in Fig. 4.

1) Chikusei Dataset: The Chikusei dataset was acquired using visible and near-infrared (NIR) imaging sensors, covering agricultural and urban areas in Tsukuba, Ibaraki Prefecture, Japan. It includes 128 spectral bands spanning the wavelength range of 363–1018 nm, with a spatial size of 2517 × 2335 pixels. For experimental convenience, the black borders in the spatial domain were removed, and the central 2048 × 2048 region was cropped for training and testing.

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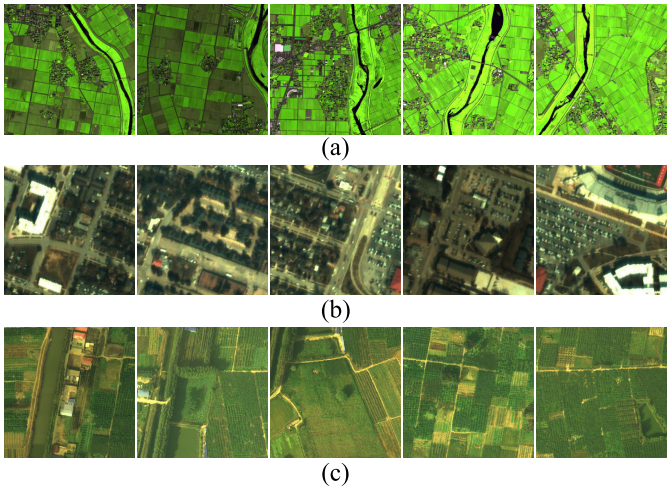


Fig. 4. Examples of the testing images selected from (a) Chikusei datasets, (b) Houston datasets, and (c) Xiongan datasets.

2) Houston Dataset: The Houston dataset originates from the 2018 IEEE GRSS Data Fusion Contest and contains hyperspectral data collected in February 2017 over the University of Houston campus. This dataset was captured using the ITRES CASI-1500 sensor, with a spatial size of 601×2384 pixels, and also includes LiDAR data obtained from the Optech Titam multiband sensor. The hyperspectral data consist of 50 spectral bands covering the wavelength range from 380 to 1050 nm.

3) Xiongan Dataset: The Xiongan dataset was acquired using a full-spectrum multimodal imaging spectrometer developed by Shanghai Institute of Technical Physics, Chinese Academy of Sciences. The spectral range is 400–1000 nm with 256 spectral bands, and the spatial size of the images is 3750×1580 pixels.

In the simulation experiments, the original HSI from the above datasets was used as the reference image. To generate the test HSI, the reference image was first blurred using a Gaussian kernel and then downsampled by a certain factor. To simulate the PAN image, the reference image was downsampled along the spectral dimension, and the spectral channels were averaged to obtain the PAN image. To generate the HR-MSI, the HR-HSI is first spectrally downsampled using a sub-spectral sampling matrix constructed based on the Nikon D700 camera's response function,[1] followed by Gaussian kernel blurring and proportional spatial downsampling of the result.

For all simulated datasets, a Gaussian filter with a standard deviation of 2 was applied to blur the reference image. Taking the Chikusei dataset as an example, the training set comprises PANs sized $16 \times 16 \times 1$, MSIs sized $4 \times 4 \times 4$, and HSIs sized $1 \times 1 \times 128$. The spatial resolutions of the panchromatic images for the datasets are as follows: 2.5 m for the Chikusei dataset, 1.0 m for the Houston dataset, and 0.5 m for the Xiongan dataset. The spatial resolutions of the MSIs and HSIs can be inferred accordingly based on these values.

[1]https://www.maxmax.com/spectral_response.htm

B. Quantitative Indicators

When evaluating the quality of HSI fusion, it is crucial to employ a comprehensive and complementary set of quantitative metrics. The six metrics adopted in this study systematically assess the results from two core dimensions: spatial structure preservation and spectral information fidelity.

1) Spatial Quality Metrics: Mean peak signal-to-noise ratio (MPSNR) and root mean square error (RMSE): both are classic pixel-error-based metrics. They directly quantify the overall closeness of the reconstructed image to the reference image from the perspectives of signal-to-noise ratio and error magnitude, respectively. Better values (higher MPSNR and lower RMSE) indicate higher global fidelity. Universal image quality index (UIQI) and mean structural similarity index (MSSIM): these metrics simulate the human visual system by measuring the similarity in luminance, contrast, and structure between images to evaluate the preservation of spatial details and texture. Higher values indicate better perceptual quality, effectively compensating for the shortcomings of pixel-error metrics in assessing perceptual consistency.

2) Spectral Quality Metrics: The spectral quality is assessed using the following two metrics:

1) *Spectral Angle Mapper (SAM):* This metric evaluates spectral shape fidelity by calculating the angle between pixel spectral vectors. A lower value indicates less spectral distortion in the fused result, which is crucial for applications relying on spectral characteristics, such as land cover classification and target detection.
2) *Relative Dimensionless Global Error in Synthesis (ERGAS):* This metric synthesizes the relative spatial and spectral errors, providing a global reflection of the overall quality of the fusion result. A lower value indicates higher overall accuracy, making it particularly suitable for assessing the fusion quality of multiresolution and multiband data.

In summary, this set of metrics forms a multidimensional evaluation system encompassing pixel error, perceptual similarity, and spectral fidelity. It systematically and complementarily verifies the proposed method's ability to effectively enhance spatial details while protecting spectral information, providing a solid quantitative basis for the comprehensiveness and effectiveness of the method.

C. Experimental Results

1) Chikusei Dataset: As shown in Table I, the proposed method outperforms all competing methods in terms of the evaluation metrics. Fig. 5 presents the pseudo-color images generated by the proposed method and several representative baseline approaches, constructed using the 70th, 100th, and 36th spectral bands. It can be observed that the HSIs reconstructed by EXP and DRCNN suffer from noticeable edge blurring, while the images obtained by HySure and FUSE exhibit poor detail restoration. The D-UNet and PSRT methods produce severe spectral distortions, and the HyperPNN method shows certain spectral distortions on some targets. In contrast, the HSIs generated by the proposed method demonstrate superior preservation of spatial details and spectral

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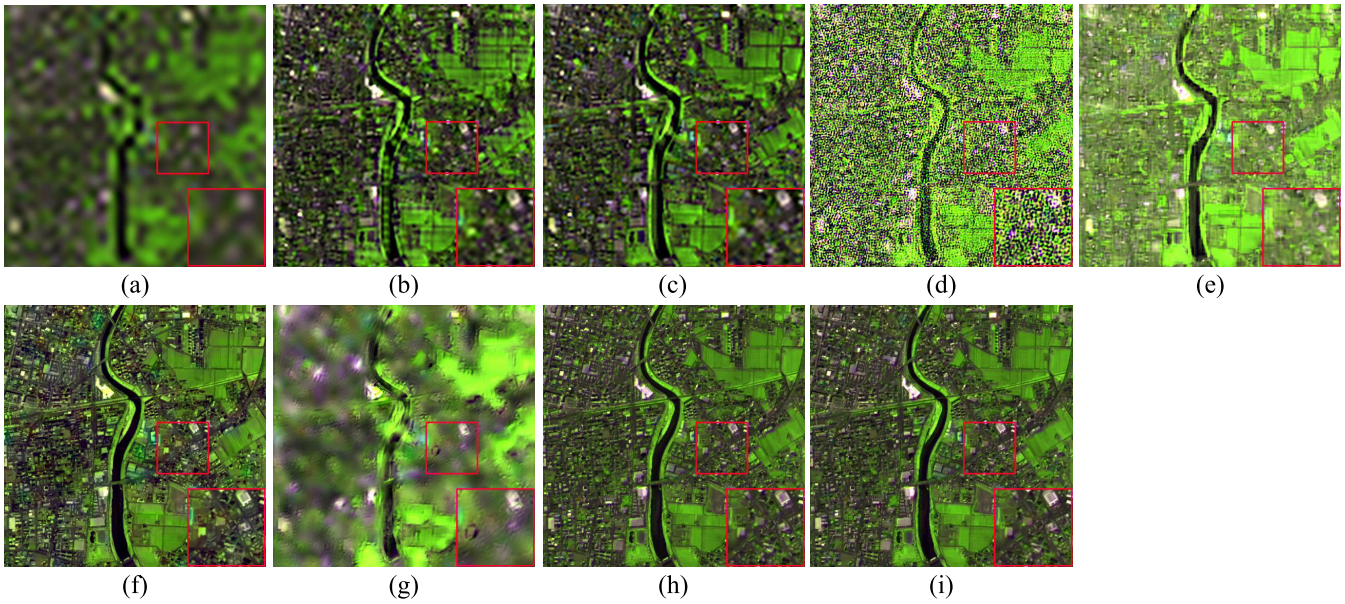


Fig. 5. Fusion images of the pseudo-color composite of the Chikusei dataset (70th, 100th, and 36th bands). The fusion is performed using (a) EXP, (b) HySure, (c) FUSE, (d) D-UNet, (e) PSRT, (f) HyperPNN, (g) DRCNN, (h) HMPNet, and (i) DIM-HMPF methods.

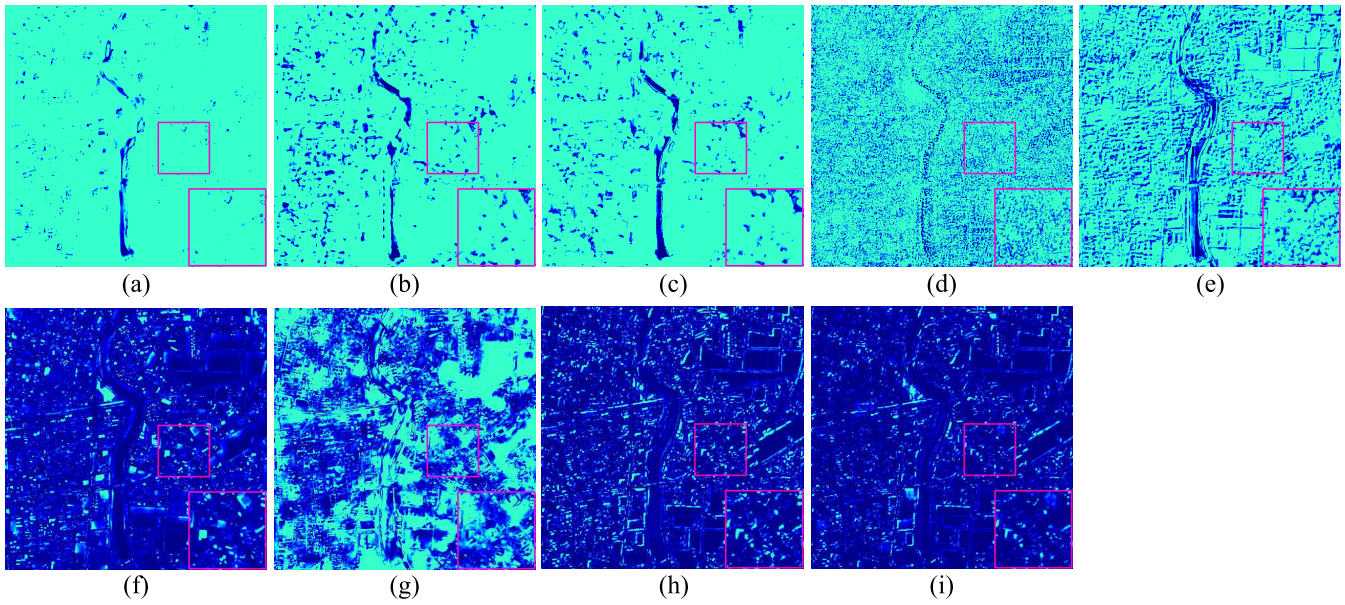


Fig. 6. Error images of the fusion of the Chikusei dataset. The fusion is performed using (a) EXP, (b) HySure, (c) FUSE, (d) D-UNet, (e) PSRT, (f) HyperPNN, (g) DRCNN, (h) HMPNet, and (i) DIM-HMPF methods.

consistency, achieving results comparable to or even better than the latest state-of-the-art methods. To provide a more intuitive comparison between the reconstructed HSIs and the reference image, Fig. 6 illustrates the mean difference maps of different methods. As shown in the figure, the results of EXP, HySure, FUSE, D-UNet, PSRT, and DRCNN deviate significantly from the reference image. The HyperPNN method exhibits poor reconstruction performance on some large targets, while HMPNet shows inferior performance in terms of detail reconstruction. In contrast, the proposed method yields much smaller differences, indicating superior reconstruction reliability. Considering that HSIs contain a large number of spectral bands, a bandwise performance analysis is further conducted, as shown in Fig. 7. From Fig. 7(a) and (b), it can be seen that the proposed method consistently achieves relatively high PSNR and SSIM values across all spectral bands, demonstrating its strong reconstruction capability. Moreover, Fig. 7(c) shows that the proposed method maintains consistently lower SAM values, indicating that it can more effectively suppress spectral distortion compared with the other competing approaches.

2) Houston Dataset: As shown in Table II, the proposed method achieves superior performance over all comparative approaches across all evaluation metrics. Fig. 8 presents pseudo-color images reconstructed by the proposed method

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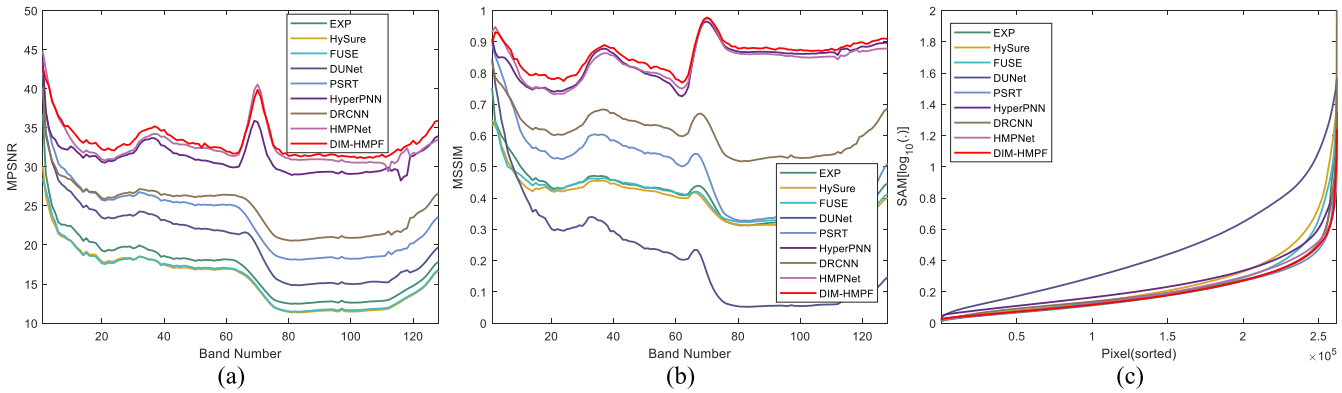


Fig. 7. Different methods were compared on the Chikusei dataset, including (a) corresponding curves of MPSNR, (b) MSSIM, and (c) SAM.

TABLE I

EVALUATION OF THE PROPOSED AND COMPARATIVE METHODS WAS CONDUCTED ON THE CHIKUSEI DATASET. THE TABLE PRESENTS THE MEAN VALUES OF MPSNR, RMSE, ERGAS, SAM, UIQI, AND MSSIM METRICS, WHERE THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Indices	EXP	HySure	FUSE	D-UNet	PSRT	HyperPNN	DRCNN	HMPNet	DIM-HMPF
MPSNR↑	19.8148	18.6587	19.0967	21.2458	24.4706	29.8852	24.5727	<u>30.4856</u>	**30.7108**
RMSE↓	38.2580	41.2853	40.2969	29.9014	20.5654	9.0432	20.5057	<u>8.6232</u>	**8.5030**
ERGAS↓	31.8133	37.5539	36.4347	12.3685	8.7386	6.0435	8.4344	<u>5.5363</u>	**5.4223**
SAM↓	7.8233	10.1042	8.5295	17.1766	<u>7.6760</u>	9.3155	8.0328	7.9084	**7.5218**
UIQI↑	0.2659	0.3897	0.4335	0.1201	0.3623	0.5573	0.2988	<u>0.5995</u>	**0.6298**
MSSIM↑	0.5415	0.5501	0.5843	0.2876	0.5995	<u>0.8316</u>	0.6570	0.8263	**0.8370**

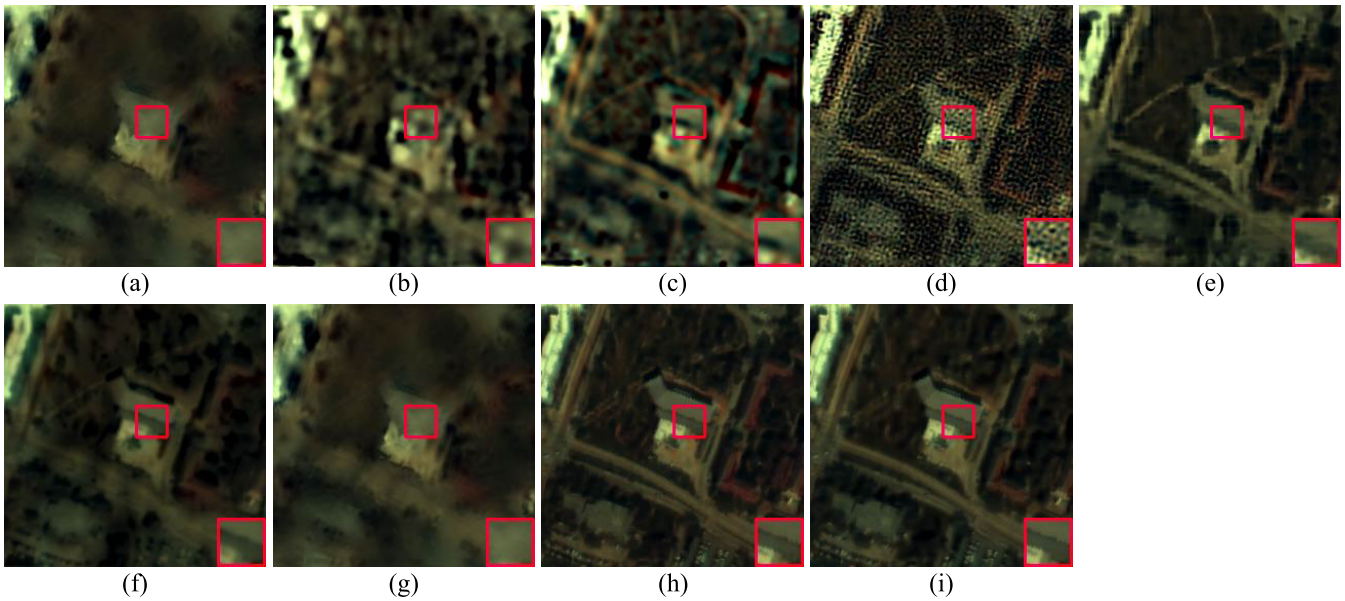


Fig. 8. Fusion images of the pseudo-color composite of the Houston dataset (20th, 10th, and 5th bands). The fusion is performed using (a) EXP, (b) HySure, (c) FUSE, (d) D-UNet, (e) PSRT, (f) HyperPNN, (g) DRCNN, (h) HMPNet, and (i) DIM-HMPF methods.

and several representative baselines using the 20th, 10th, and 5th spectral bands. It can be observed that the HSIs reconstructed by EXP and DRCNN contain almost no discernible spatial details. The images recovered by HySure, FUSE, and HyperPNN appear overly blurred with significant detail loss, while the D-UNet method introduces considerable noise into the reconstruction. In contrast, PSRT, HMPNet, and the proposed DIM-HMPF method produce the best overall reconstruction results, with the proposed method showing particularly strong performance in terms of spatial detail preservation and visual consistency. To more intuitively compare the reconstructed HSIs with the reference image, Fig. 9

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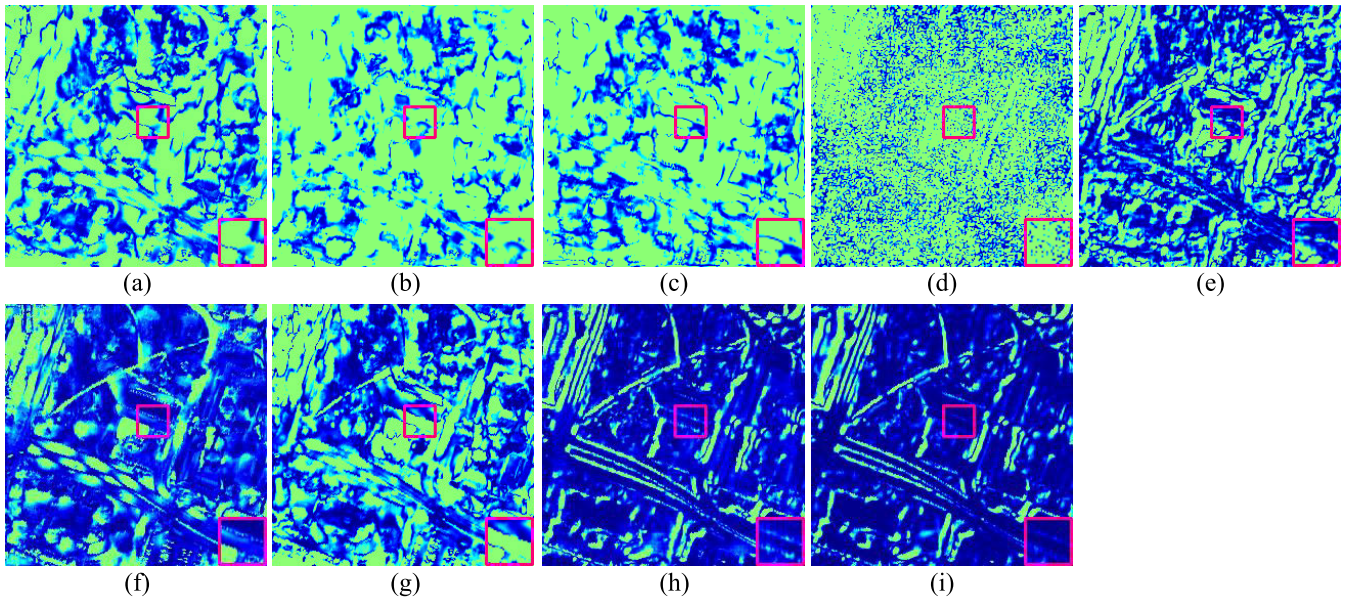


Fig. 9. Error images of the fusion of the Houston dataset. The fusion is performed using (a) EXP, (b) HySure, (c) FUSE, (d) D-UNet, (e) PSRT, (f) HyperPNN, (g) DRCNN, (h) HMPNet, and (i) DIM-HMPF methods.

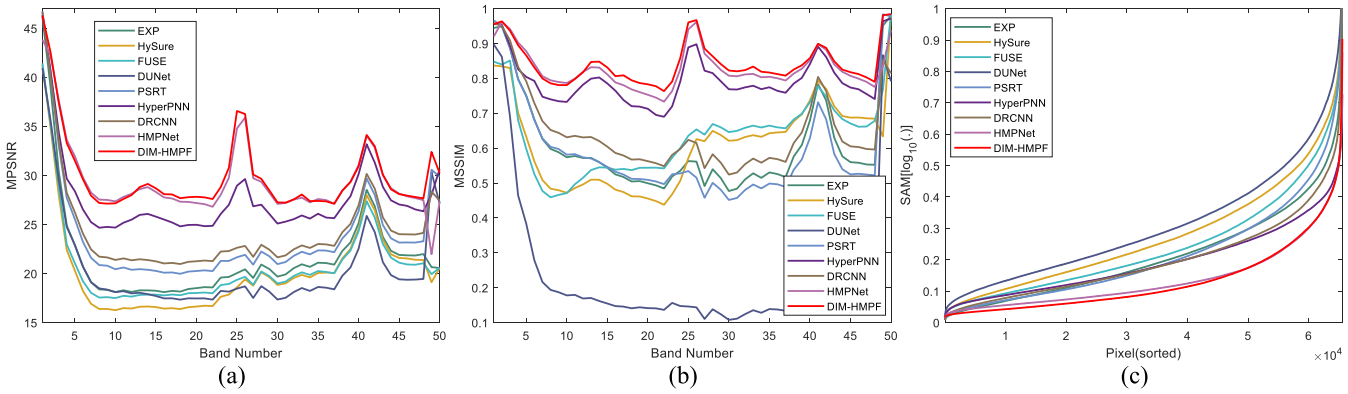


Fig. 10. Different methods were compared on the Houston dataset, including (a) corresponding curves of MPSNR, (b) MSSIM, and (c) SAM.

TABLE II

EVALUATION OF THE PROPOSED AND COMPARATIVE METHODS WAS CONDUCTED ON THE HOUSTON DATASET. THE TABLE PRESENTS THE MEAN VALUES OF MPSNR, RMSE, ERGAS, SAM, UIQI, AND MSSIM METRICS, WHERE THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Indices	EXP	HySure	FUSE	D-UNet	PSRT	HyperPNN	DRCNN	HMPNet	DIM-HMPF
MPSNR↑	21.2089	19.6840	19.2039	20.1519	23.1754	27.2662	23.1472	<u>29.5081</u>	**29.6847**
RMSE↓	27.0526	31.4458	33.1764	29.2624	21.2701	13.6048	20.8107	<u>10.7359</u>	**10.1448**
ERGAS↓	15.2350	18.1299	19.2215	8.2095	5.8848	3.6519	5.9206	<u>2.8903</u>	**2.8216**
SAM↓	11.6543	14.8673	15.6651	19.2463	12.6434	12.4985	11.6183	<u>8.3288</u>	**8.1286**
UIQI↑	0.3775	0.4487	0.4648	0.2635	0.4870	0.7117	0.3866	<u>0.7875</u>	**0.7921**
MSSIM↑	0.5824	0.5969	0.5374	0.2327	0.5398	0.7987	0.6295	<u>0.8461</u>	**0.8519**

provides the mean difference maps for the various methods. As shown, the results produced by EXP, HySure, and FUSE exhibit clear deviations from the reference image; D-UNet introduces substantial noise-related errors; PSRT and DRCNN still display a noticeable amount of residual error; HyperPNN suffers from larger deviations on several large-scale targets; and HMPNet retains certain inaccuracies in fine-detail reconstruction. In contrast, the proposed method yields significantly smaller differences, demonstrating superior fidelity and reconstruction reliability. Considering that HSIs contain a large number of spectral bands, a bandwise performance evaluation was further conducted, as illustrated in Fig. 10. From Fig. 10(a) and (b), it is evident that the proposed method consistently achieves the highest PSNR and SSIM

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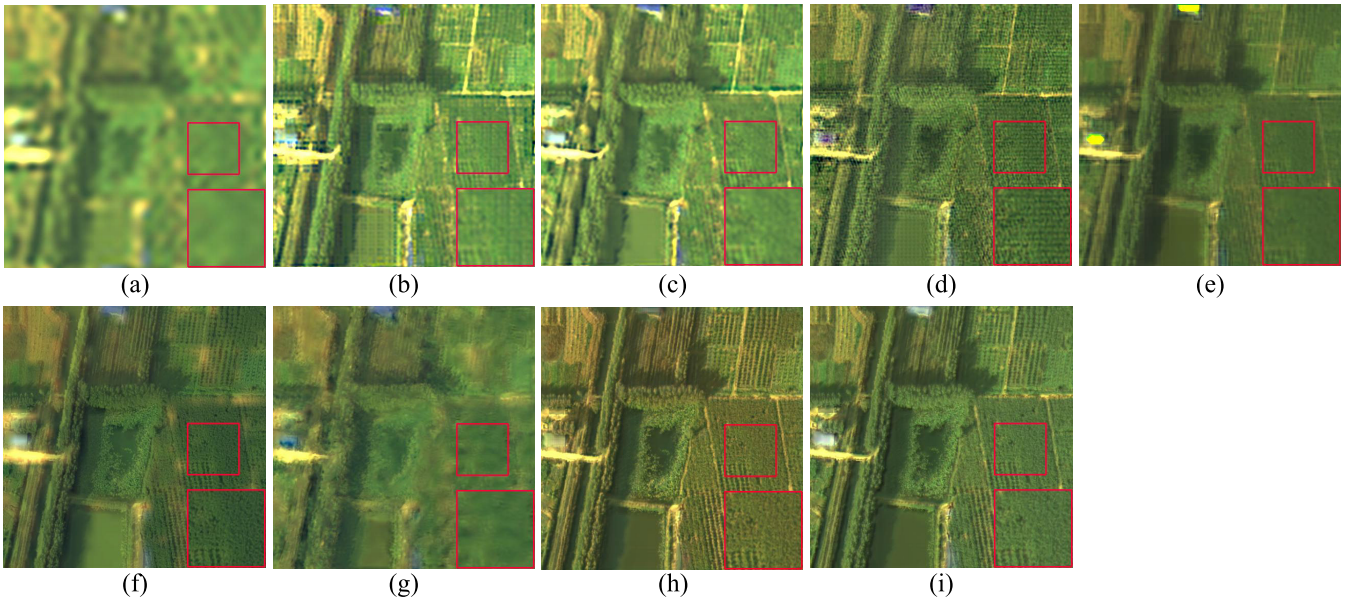


Fig. 11. Fusion images of the pseudo-color composite of the Xiongan dataset (108th, 68th, and 24th bands). The fusion is performed using (a) EXP, (b) HySure, (c) FUSE, (d) D-UNet, (e) PSRT, (f) HyperPNN, (g) DRCNN, (h) HMPNet, and (i) DIM-HMPF methods.

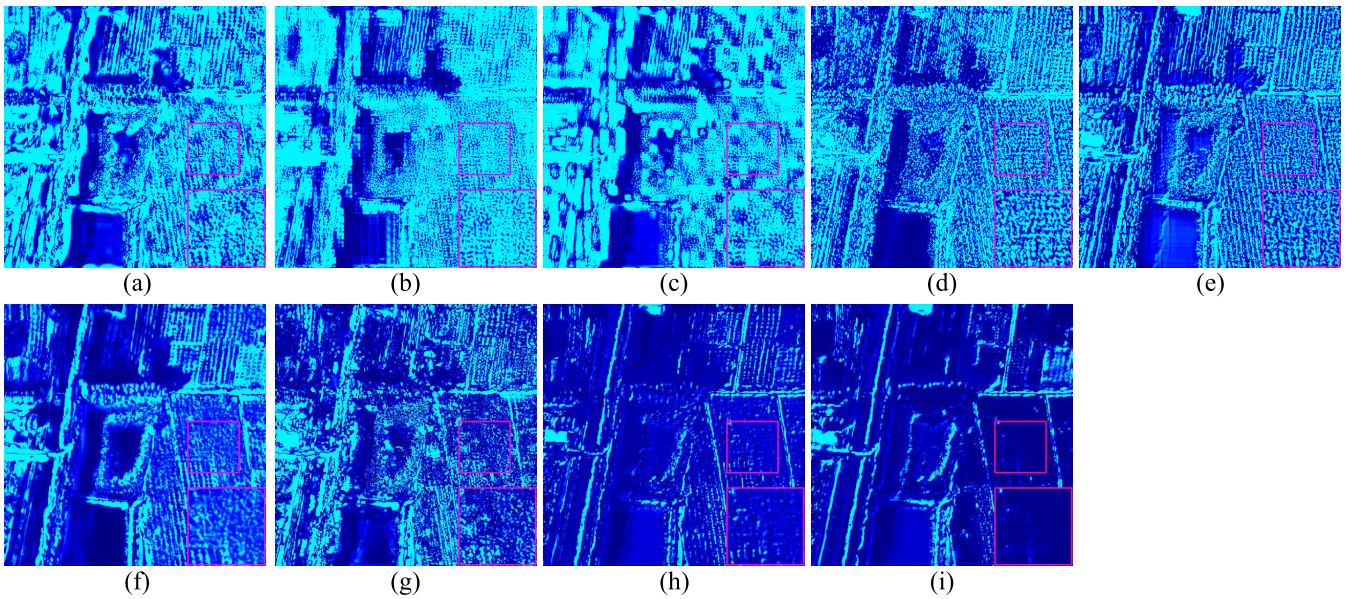


Fig. 12. Error images of the fusion of the Xiongan dataset. The fusion is performed using (a) EXP, (b) HySure, (c) FUSE, (d) D-UNet, (e) PSRT, (f) HyperPNN, (g) DRCNN, (h) HMPNet, and (i) DIM-HMPF methods.

values across all spectral bands, highlighting its strong and stable reconstruction capability. In addition, Fig. 10(c) shows that the proposed method maintains consistently lower SAM values, indicating that it suppresses spectral distortion more effectively than the competing approaches.

3) Xiongan Dataset: As shown in Table III, the proposed method outperforms all comparative methods across all evaluation metrics. Fig. 11 presents pseudo-color images reconstructed by the proposed method and several representative baseline approaches using the 108th, 68th, and 24th spectral bands. It can be observed that the HSIs reconstructed by EXP and DRCNN almost completely lose discernible spatial details. The images recovered by HySure, FUSE, D-UNet, PSRT, and HyperPNN are generally blurred with insufficient spatial detail. In contrast, HMPNet and the proposed DIM-HMPF method achieve the best reconstruction performance, with the proposed method showing superior detail recovery and visual consistency. To further visually compare the reconstructed HSIs with the reference image, Fig. 12 illustrates the mean difference maps of different methods. As shown in the figure, the results produced by EXP, HySure, FUSE, D-UNet, and PSRT exhibit significant deviations from the reference image. HyperPNN and DRCNN perform poorly in terms of fine-detail reconstruction, while HMPNet shows inferior reconstruction quality in the paddy field regions. In contrast, the proposed method yields the smallest differences,

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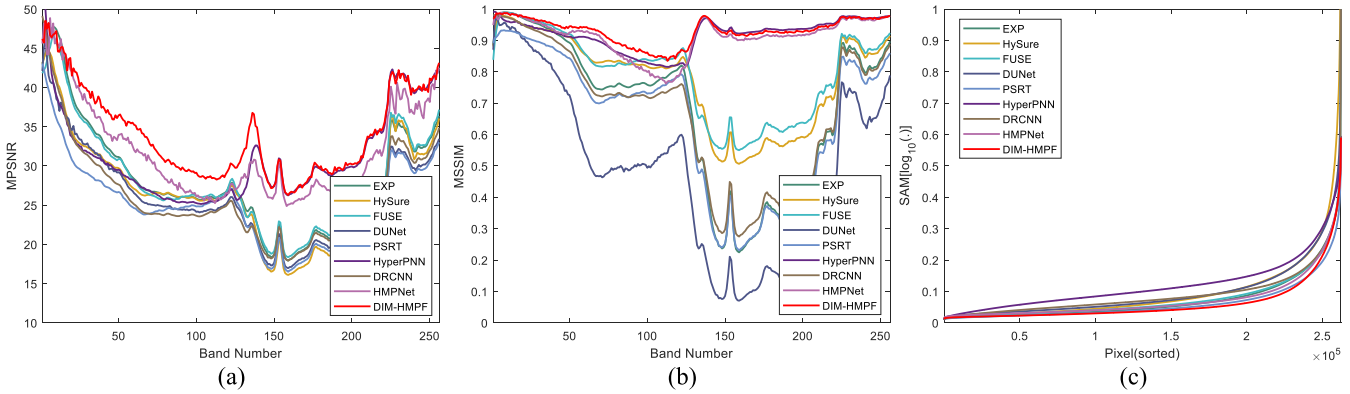


Fig. 13. Different methods were compared on the Xiongan dataset, including (a) corresponding curves of MPSNR, (b) MSSIM, and (c) SAM.

TABLE III

EVALUATION OF THE PROPOSED AND COMPARATIVE METHODS WAS CONDUCTED ON THE XIONGAN DATASET. THE TABLE PRESENTS THE MEAN VALUES OF MPSNR, RMSE, ERGAS, SAM, UIQI, AND MSSIM METRICS, WHERE THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Indices	EXP	HySure	FUSE	D-UNet	PSRT	HyperPNN	DRCNN	HMPNet	DIM-HMPF
MPSNR↑	24.5808	24.2557	24.3028	26.3788	27.5251	30.2098	27.7086	<u>31.1088</u>	**31.6778**
RMSE↓	22.2211	22.5037	22.4757	17.2967	14.9757	10.4061	14.6799	<u>9.5286</u>	**8.3853**
ERGAS↓	9.2583	9.5382	9.5698	3.5537	3.1502	2.5358	2.9791	<u>2.2804</u>	**2.0973**
SAM↓	4.6969	5.5698	5.0991	6.1027	4.3775	6.6465	<u>5.2239</u>	5.2637	**4.4631**
UIQI↑	0.2514	0.5918	0.5454	0.1457	0.3188	0.7171	0.2079	<u>0.7358</u>	**0.7403**
MSSIM↑	0.6589	0.7595	0.7606	0.4853	0.6450	<u>0.9003</u>	0.6727	0.8859	**0.9020**



Fig. 14. Visualization of intermediate features, (70th, 100th, and 36th bands) of feature cubes (a) $\mathcal{X}^1$, (b) $\mathcal{X}^2$, (c) $\mathcal{X}^3$, (d) $\mathcal{X}^4$, and (e) $\mathcal{X}^5$.

demonstrating superior reconstruction fidelity and reliability. Considering that HSIs contain a large number of spectral bands, a bandwise performance analysis is further conducted, as shown in Fig. 13. From Fig. 13(a) and (b), it is evident that the proposed method consistently achieves the highest PSNR and SSIM values across all spectral bands, highlighting its strong and stable reconstruction capability. Moreover, Fig. 13(c) shows that the proposed method maintains consistently lower SAM values, indicating that it suppresses spectral distortion more effectively than the competing approaches.

The experimental results on the Chikusei, Houston, and Xiongan datasets consistently demonstrate that the proposed method achieves stable and superior reconstruction performance across diverse and complex scenarios. Quantitatively, the proposed method outperforms or significantly surpasses the competing approaches in spatial reconstruction metrics such as PSNR and SSIM, as well as in spectral fidelity metrics such as SAM. Qualitatively, it effectively suppresses noise, reduces blurring and artifacts, and simultaneously preserves spatial details and spectral consistency. Bandwise analysis further indicates that the method maintains strong and stable reconstruction capability across all spectral dimensions, fully validating its robustness, generalization ability, and potential for practical applications under multiscene and varied data distribution conditions.

D. Parameter Analysis

1) Visualization of Different Stages: Using the test image from the Chikusei dataset (as shown in Fig. 14), we illustrate the evolution of intermediate HSIs $\mathcal{X}^t$ generated by DIM-HMPF at different stages. The figure presents the inferred

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TABLE IV
THIS STUDY AIMS TO EXPLORE THE IMPACT OF VARYING THE NUMBER OF ITERATIONS ON THE NETWORK PERFORMANCE FOR THE ICVL DATASETS

T	PSNR	RMSE	ERGAS	SAM	UIQI	SSIM
1	30.4717	8.6537	5.5946	7.7520	0.6270	0.8327
3	30.1726	8.6559	5.7823	8.5711	0.6260	0.8301
5	**30.7108**	**8.5030**	**5.4223**	**7.5218**	**0.6298**	**0.8370**
7	30.2065	8.5209	5.9489	10.0520	0.6279	0.8180

results at each stage, showing that as the processing progresses, the fusion results are progressively enhanced in both spatial detail and spectral information. In the early stages, the images may exhibit slight blurring or spectral distortion; however, as the network iterates and feature fusion proceeds, edges and textures become sharper, and spectral consistency is effectively improved. This visualization provides an intuitive demonstration of how DIM-HMPF gradually refines image quality through multiple stages, offering direct evidence of its effectiveness in HSI fusion tasks.

2) Setting of the Iteration Number T in the Network: The DIM-HMPF network consists of N iterative stages, with each stage progressively refining the spatial details and spectral consistency of the HSIs through recursive optimization. The setting of the parameter T plays a critical role in the network's performance. We systematically evaluated DIM-HMPF under different T settings, as summarized in Table IV. The results indicate that increasing the number of iterative stages significantly enhances the network's representational capacity and fusion quality, leading to sharper edges and improved spectral fidelity. However, an excessively high number of iterations (e.g., T = 7) may also lead to overfitting, which in turn can cause performance fluctuations or degradation, while significantly increasing computational cost and runtime. Therefore, a tradeoff between performance and efficiency must be considered. As shown in Table IV, setting $T = 5$ achieves an optimal balance between reconstruction performance and computational efficiency. Therefore, we adopt $T = 5$ as the number of iterative stages in our experiments.

V. CONCLUSION

The core of fusing HSI, MSI, and PAN lies in extracting spatial details from MSI and PAN and injecting them into HSI to generate HR-HSIs. However, existing methods largely rely on black-box DL models and lack designs grounded in this fundamental principle, resulting in limited physical interpretability. To address this limitation, we propose DIM-HMPF, a deep network for the fusion of HSI–MSI–PAN using a deep unfolding approach. This innovative network effectively injects the spatial details of MSI and PAN into HSI while preserving its spectral information. We utilize a physical model of detail injection for unfolding and employ DL to learn this process. In addition, we adopt an optimization problem-solving approach based on the proximal gradient descent (PGD) algorithm, implemented through a series of iterative steps. By unfolding these iterative steps into several network modules, we achieve the final DIM-HMPF model. The experimental results on simulated datasets qualitatively and quantitatively demonstrate the superiority of DIM-HMPF over other state-of-the-art methods.

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